VIOLA DE RUVO

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Research Interests

Computational and theoretical neuroscience, machine learning, and statistical physics. My current work combines data-driven modelling of neural recordings with theory to understand how much of a neural circuit's computation can be explained by random synaptic connectivity, and where structured connectivity and plasticity are required.

Education

Bocconi University Milan, Italy
M.Sc. in Artificial Intelligence *2023 – 2026*

- Advisor: Prof. Nicolas Brunel
- Thesis: "Cerebellar Computation Under Random and Structured Synaptic Connectivity." Fitted a feedforward rate model of the cerebellar circuit to neural recordings from behaving monkeys, to determine where random connectivity suffices and where structure is required.

Sapienza University of Rome Rome, Italy
B.Sc. in Physics *2019 – 2023*

- Advisor: Prof. Rinaldo Trotta
- Thesis: "Foundations of Interaction-Free Measurement." Analyzed the core ideas behind interaction-free measurement and their realization in interferometric experiments.

Publications

Conference Proceedings

- **De Ruvo, V.**, Muti, A., Dementieva, D., & Nozza, D. (2025). Detoxify-IT: An Italian Parallel Dataset for Text Detoxification. In *Proceedings of the 9th Workshop on Online Abuse and Harms (WOAH)* (pp. 267–275).

In Preparation

- **De Ruvo, V.**, Kuśmierz, Ł., & Brunel, N. Cerebellar computation under random and structured synaptic connectivity. *Manuscript in preparation.*

Invited Talks

- (Feb 2026) NeuroML Journal Club, Bocconi University, Milan, Italy [Talk] **De Ruvo, V.** "Evaluating Random Connectivity in the Cerebellar Cortex"

Research Experience

Bocconi University, Department of Computing Sciences Milan, Italy
Research Assistant — Prof. Nicolas Brunel *Feb 2025 – Present*

- Data-driven modelling of the cerebellar circuit during smooth-pursuit eye movements in monkeys.
- Analyzed population-level neural activity from electrophysiological recordings.
- Developed an empirically constrained generative model of the circuit's inputs.
- Developed and fitted a feedforward rate model of the cerebellar circuit to recorded activity.
- Compared random and optimized synaptic weights, identifying which properties of the recorded activity random connectivity fails to reproduce.
- Investigated biologically plausible learning rules that drive plasticity in the circuit.

Bocconi University, Natural Language Processing Lab Milan, Italy
Research Assistant — Prof. Debora Nozza *Oct 2024 – Apr 2025*

- Italian text detoxification: rewriting toxic comments into non-toxic text while preserving their meaning.
- Built Detoxify-IT, the first Italian parallel detoxification dataset, pairing toxic comments with human-written neutral rewrites.

- Compared multilingual models trained on other languages, zero-shot prompted LLMs, and models fine-tuned on Italian data; showed that even limited Italian fine-tuning outperforms the alternatives. Published at WOAH 2025.

Volunteering

Scomodo – **Magazine** Rome, Italy
Contributor *2015 – 2020*

- Independent magazine and cultural community run entirely by students and young people under 30.
- Wrote articles / helped organize cultural events / distributed the magazine in schools and universities.

Libera – **Association against mafia** Rome, Italy
Volunteer *2014 – 2017*

- Helped run awareness activities on legality and anti-mafia civic engagement.

Technical Skills

Languages: Python, C/C++, Matlab, R
ML Libraries: PyTorch, scikit-learn
Developer Tools: Git, GitHub
Spoken Languages: Italian (native), English (fluent)

Relevant Coursework

- Computational Neuroscience
- Complex Systems and Physical Models
- Bayesian Statistical Methods
- Deep Learning and Reinforcement Learning
- Algorithms for Optimization and Inference
- Information Theory
- Statistical Mechanics
- Quantum Mechanics
- Computational Physics
- Signals and Systems